

AMALIY MASHG'ULOT-11

Mavzu: C++da shartni oldin tekshiruvchi operatorlar. Takrorlanuvchi strukturali dasturlar. C++da keyin tekshiruvchi operatorlar. Takrorlanuvchi strukturali dasturlar.

I-topshiriq

amaliy mashg'ulotning maqsadi: Talabalarda C++ Builderda murakkab sikllarni ishlatishni va natija olishni o'rgatish.

a) Ichma-ich joylashdan sikllardan foydalanib ifoda qiymatini hisoblash dasturini tuzing:

$$S = \sum_{k=1}^{15} \sum_{n=1}^{10} (a^k + x^n)$$

1) Masalani yechish dasturi.

```
#include<iostream>
using namespace std;
int main(){
float S=0,P;
for(int k=1;k<=15;k++){
P=0;
for (int n=1; n<=10;n++) {
P+=(pow(a,k)+pow(x,n));
} S+=P;
}
cout<<"S="<<S;
return 0;
}
```

4-amaliy mashg'ulot topshiriqlari

a) ichma-ich joylashgan sikllardan foydalanib dastur tuzing;

amaliy mashg'ulot topshiriqlarni bajarishda har bir talaba dastlab takrorlanish buyruqlari haqida qisqacha ma'lumot berib, so'ngra berilgan amaliy topshiriqlarni bajaradi.

1-variant

a) $S = \sum_{n=1}^5 \sum_{i=1}^{12} (n^2 + i^3)$ ni hisoblang

2-variant

a) $S = \prod_{a=1}^{10} \prod_{i=2}^4 \frac{k^3}{k+a}$ ni hisoblang.

3-variant

a) $S = \sum_{n=1}^5 \prod_{i=1}^8 (n^2 + i^3)$ ni hisoblang

4-variant

a) $S = \prod_{k=1}^5 \sum_{j=1}^{10} (k^2 + j^4)$ ni hisoblang

5-variant

$$S = \sum_{n=1}^3 \prod_{k=1}^5 \sum_{j=1}^{10} (k^2 + j^2 + n)$$

a) ni hisoblang.

6-variant

$$S = \prod_{k=1}^4 \sum_{j=1}^{10} (k! + j!)$$

a) ni hisoblang

7-variant

$$S = \sum_{n=1}^3 \prod_{k=1}^4 \sum_{j=1}^5 (k! + j! + n!)$$

a) ni hisoblang.

8-variant

$$S = \sum_{k=1}^3 \left(\prod_{n=1}^4 kn^2 + \sum_{i=1}^5 ki^3 \right)$$

a) ni hisoblang.

9-variant

$$S = \prod_{k=1}^3 \left(a \prod_{n=1}^4 kn^2 + b \sum_{i=1}^5 ki^3 \right)$$

a) ni hisoblang

10-variant

$$S = \prod_{k=1}^4 \sum_{j=1}^{10} \frac{(k! + j!)}{a + b}$$

a) ni hisoblang.

11-variant

$$S = \prod_{k=1}^4 \sum_{j=1}^{10} \frac{a(k + j)}{(k! + j!)}$$

a) ni hisoblang

12-variant

$$S = \prod_{k=1}^4 \frac{\sum_{j=1}^{10} j!}{k!}$$

a) ni hisoblang.

13-variant

$$S = \sum_{i=1}^3 \frac{\left(\sum_{j=1}^{10} j + \prod_{k=1}^4 \frac{1}{k} \right)}{i!}$$

a) ni hisoblang.

14-variant

a) Ko'paytirish jadvali elementlarini hosil qilish dasturini tuzing.

15-variant

$$S = \sum_{n=1}^{10} \prod_{k=1}^5 (a^k + b^n)$$

a) ni hisoblang.

16-variant

$$S = \prod_{n=1}^3 \prod_{k=1}^5 (x^k + y^n)$$

a) ni hisoblang.

17-variant

$$S = \sum_{n=1}^5 \prod_{k=1}^5 (x^k + y^n)$$

a) ni hisoblang.

18-variant

$$P = \prod_{a=1}^5 \prod_{k=2}^4 \frac{b^2}{k + a}$$

a) ni hisoblang.

